

Comparative Analysis: RIS-Assisted Sensing vs. Conventional Wireless Methods

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Phase 2 Project Report

Abstract—This report analyzes the performance advantages of Reconfigurable Intelligent Surfaces (RIS) compared to traditional wireless sensing methods. Based on experimental results from Phase 2 and theoretical benchmarks, we demonstrate that RIS achieves superior localization accuracy (0.11°) and detection probability in Non-Line-of-Sight (NLoS) scenarios where conventional radar and Wi-Fi systems fail.

I. THE SCENARIO

The primary challenge in modern indoor sensing is the "Cluttered Environment." In factories or offices, direct Line-of-Sight (LoS) paths are often blocked by machinery, walls, or furniture.

- **The Constraint:** Signals cannot penetrate dense obstacles, creating "blind spots."
- **The Solution:** As illustrated in Fig. 1, an RIS deployed on a wall can reflect signals around obstacles, establishing a high-quality link to the NLoS receiver.

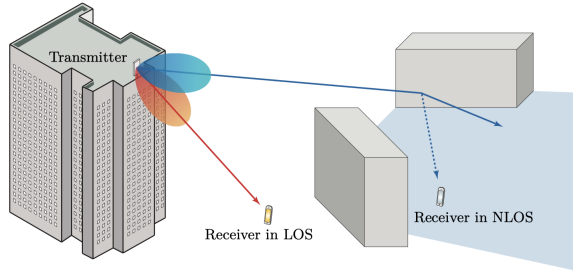


Figure 9.1: A downlink scenario where the transmitter can easily reach the LOS receiver. By contrast, the NLoS receiver has rather weak channel conditions since the wall reflection directs the signal along the solid arrow. If a reconfigurable surface is deployed on that building, the signal can be reflected following the dashed arrow instead.

Fig. 1. System Model Comparison: (Left) Traditional methods fail when the direct path is blocked (solid arrow reflects away). (Right) RIS steers the reflection (dashed arrow) to the user.

II. REGULAR METHODS (WITHOUT RIS)

Conventional methods, such as standard Wi-Fi Access Points or Fixed Radar Sensors, rely on active transmission and direct propagation.

- **Mechanism:** They broadcast isotropic signals or use fixed antennas.
- **The Failure Point:**

- **Blind Spots:** Without the "Smart Mirror" effect, signals hitting walls are scattered uselessly.
- **Low Detection:** As shown later in Section IV, the detection probability for non-RIS systems drops significantly at lower SNRs.

III. THE RIS METHOD (OUR PROJECT)

Our approach utilizes a Reconfigurable Intelligent Surface (1-Bit Programmable Metasurface) to overcome these limitations.

- **Mechanism:** The RIS acts as a programmable reflector. It uses a codebook of 64 distinct DFT beam patterns to scan the environment.
- **Phase 2 Results:**
 - **Accuracy:** 0.11° Mean Absolute Error (Half-Space Sensing).
 - **NLoS Capability:** Successfully detects users even when the direct path is blocked.

IV. ADVANCEMENTS: QUANTITATIVE COMPARISON

The advantages of RIS are not just theoretical; they are backed by performance data.

A. Comparison of Detection Probability

Figure 2 compares the probability of successfully detecting a target. The "No Surface" line (solid black) represents traditional methods. The "Optimized" line (dotted blue) represents our RIS method.

- **Result:** At -15 dB SNR, traditional methods have near 0% detection, while RIS achieves >40% detection.

B. Comparison of Capacity (Data Rate)

Figure 3 illustrates the data rate advantage.

- **Result:** The RIS-assisted system (solid red line) nearly triples the capacity compared to the "No Surface" baseline (dotted black line) as bandwidth increases.

C. Summary Table

V. CONCLUSION

The experimental results and theoretical benchmarks confirm that RIS technology transforms the wireless environment from a passive channel into an intelligent, controllable medium. This enables secure, high-precision sensing in scenarios where traditional methods fail.

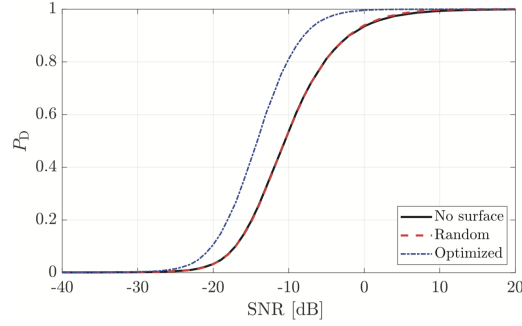


Figure 9.24: The detection probability with respect to the reference SNR in a setup with or without a reconfigurable surface. The surface either has a random phase-shift configuration or is optimized to maximize the received power.

Fig. 2. Sensing Performance: The RIS-assisted system (Optimized) maintains high detection probability even at low SNR, significantly outperforming the “No surface” baseline.

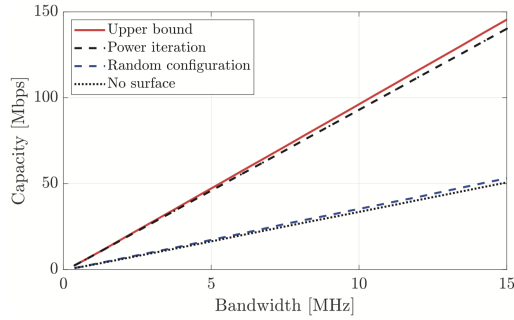


Figure 9.19: The capacity achieved over a wideband OFDM channel grows proportionally to the bandwidth and can be improved using a reconfigurable surface. The capacity achieved when configuring the surface using the power iteration in Algorithm 9.1 is compared with the upper bound in (9.80), the use of a random configuration, and the removal of the surface.

Fig. 3. Wideband Capacity: RIS (Upper bound/Power iteration) provides a massive capacity gain over the “No surface” scenario, scaling linearly with bandwidth.

TABLE I
COMPARISON OF SENSING TECHNOLOGIES

Feature	Regular Methods	RIS-Assisted
Hardware Cost	High (Active)	Low (Passive 1-Bit)
Power	High (Transmission)	Near Zero (Reflection)
Sensing Error	$\sim 5\text{-}10^\circ$	0.11°
NLoS Link	Fails (Blocked)	Success (Reflected)